BirdGEN: Conditional Spectrogram Augmentation for Bird Species Classification under Simulated Label Scarcity

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Abstract

BirdGEN investigates whether conditional spectrogram generation can improve bird-species classification under simulated label scarcity. We convert recording-isolated audio from three bird species into log-mel spectrograms and compare conditional VAE-v3 and Diffusion generators for classifier augmentation. Each newly initialized CRNN receives 50 real spectrograms per species, with or without additional generated samples. Across three real-data subset choices and three CRNN initialization seeds, validation selected 200 additional generated spectrograms per species for both generators. On a fixed 489-clip real test set, mean macro-F1 was 84.75% without augmentation, 87.48% with VAE-v3, and 86.21% with Diffusion. These results support classifier utility in the tested simulated low-resource setting, but do not establish performance on genuinely rare or unseen species.

Attestation of Teamwork

Shenghao Jin contributed to data preparation and developed and evaluated the conditional VAE. Harvey Li contributed to data preparation, the Residual CNN and CRNN classifiers, the low-resource augmentation experiment, and the VAE-Diffusion generator-only spectrogram-sampling comparison. Vincent Wang developed and evaluated the conditional Diffusion model. All members contributed to the preparation and revision of the final report.

1 Introduction and Problem Formulation

Passive acoustic monitoring can scale species detection beyond manual surveys. BirdNET demonstrates this potential with a deep neural network trained on extensive annotated bird-sound data to identify hundreds of species [1]. However, supervised systems depend on labeled examples, and ecological recordings are often imbalanced and expensive to label, leaving under-represented species with limited training data. Spectrogram augmentation offers one way to address this bottleneck. ECOGEN showed the potential of generated spectrograms for classification on real recordings, reporting a 12% average accuracy improvement [2].

Motivated by this downstream evaluation, BIRDGEN asks whether class-conditional VAE-v3 and Diffusion generators can improve classification when labeled data are scarce. To measure downstream classifier utility, we use the paired change in held-out macro-F1:

$$\Delta F1 = F1_{\text{real+generated}} - F1_{\text{real-only}},$$

where macro-F1 weights all three species equally.

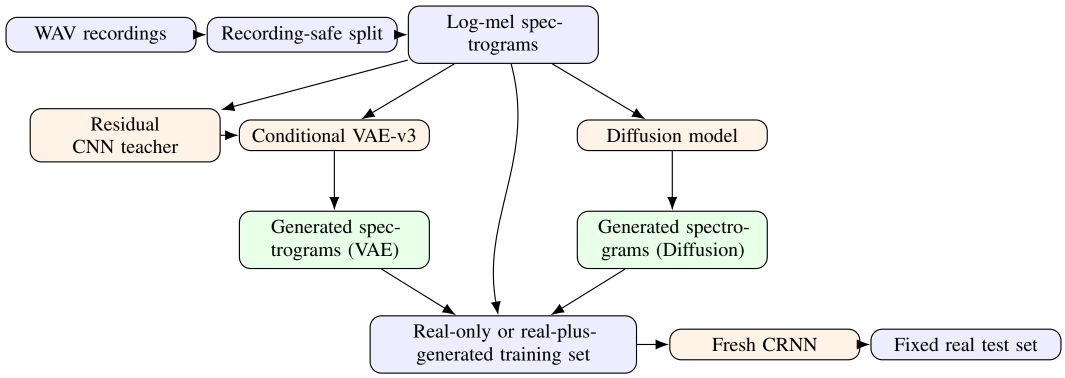

Figure 1: BirdGEN pipeline. Both generated-spectrogram branches feed the same downstream-classifier protocol.

2 Data Preparation

We use three-second, mono 22.05-kHz WAV clips from the Bird Song Data Set, compiled from Xeno-canto recordings [5]. We select American Robin, Northern Cardinal, and Song Sparrow, giving 3,347 clips from 310 source recordings. Splitting by recording ID prevents clips from the same recording from appearing across train, validation, and test sets.

Each clip is converted to a 128×128 log-mel spectrogram using a 1,024-point FFT and hop length 512. The VAE uses training-set global standardization, while the classifier maps each clip's relative 80-dB range to $[-1, 1]$; an adapter converts VAE outputs to the classifier scale.

3 System Design and Models

Figure 1 shows the matched evaluation path. Every downstream condition trains a fresh CRNN and uses the same real test set.

3.1 Classifiers

The 1.66-M-parameter Residual CNN uses six residual blocks and global average/max pooling; its frozen features and logits guide VAE-v3 training. The selected 404-k-parameter CRNN combines convolutional spectral features with a bidirectional GRU and mean/max temporal pooling [6]. The selected CRNN is frozen for generator-quality diagnostics, whereas freshly initialized CRNNs provide the downstream augmentation evidence.

3.2 Conditional VAE Evolution

We selected a conditional VAE (CVAE) rather than an unconditional VAE, because every training spectrogram is paired with a bird-species label, and the downstream task requires controllable, class-labelled augmentation [3, 4]. The species label y conditions both the encoder $q_\phi(z \mid x, y)$ and decoder $p_\theta(x \mid z, y)$. Since the requested class is supplied explicitly, the latent variables do not need to represent species identity alone and can thus focus more on within-species variation, such as vocalization type, pitch, and timing. Conditioning reduces cross-species ambiguity, although it does not by itself guarantee realistic or diverse outputs.

V1 established a CVAE baseline with a 128-dimensional global latent vector, species embeddings concatenated at the encoder and decoder, and MSE+KL loss. Its compressed bottleneck and pixel-level reconstruction objective tended to smooth narrow calls, pitch tracks, and transient structure. V2 replaced the global vector with a $16 \times 8 \times 8$ spatial latent and added residual FiLM conditioning [7], resize-convolution, and event-weighted, multiscale, and time/frequency-gradient reconstruction losses. These changes were intended to preserve the locations and edges of short transients and harmonics. For generation, V2 fitted one moment-matched diagonal Gaussian to the training posteriors of each species. Although this reduced the mismatch with standard-normal sampling, a single Gaussian could still average distinct vocalization modes within a species.

V3 increased the latent resolution to $16 \times 16 \times 16$, used a lower KL weight with a longer warmup and per-channel free bits to reduce excessive latent compression and help retain information needed for fine spectro-temporal detail. It also added feature and label consistency losses from a frozen, supervised Residual CNN species classifier trained on labeled real spectrograms, providing species-aware guidance. For generation, V3 replaced each single species-level Gaussian with a train-only posterior-anchor bank, allowing samples to be drawn near multiple encoded vocalization examples. This generation-time strategy mitigates the practical mismatch between the standard-normal prior and learned posteriors, but it does not replace the prior used to train the checkpoint. Since multiple components were modified between versions, the results show the improvement of each complete model rather than the individual causal effect of any single modification.

VAE-v3 has approximately 5.37 M parameters. Species embeddings modulate residual encoder and decoder blocks through FiLM. Its training objective is

$$\mathcal{L} = \mathcal{L}_{\text{event}} + 0.25\mathcal{L}_{\text{ms}} + 0.60\left(\mathcal{L}_{\Delta t} + \mathcal{L}_{\Delta f}\right) + \beta\mathcal{L}_{\text{KL,free}} + s_{\text{cls}}\left[0.08\mathcal{L}_{\text{feat}} + 0.03\mathcal{L}_{\text{label}}\right]. \tag{1}$$

Here, $\mathcal{L}_{\text{event}}$ is an event-weighted L1 reconstruction term, $\mathcal{L}_{\text{ms}}$ averages L1 losses at multiple resolutions, and $\mathcal{L}_{\Delta t}$ and $\mathcal{L}_{\Delta f}$ are edge-weighted L1 discrepancies between first-order time and frequency differences. The KL coefficient β is warmed up to 5×10^{-4} over 20 epochs, while s_{cls} increases to one over eight epochs. The free-bits term applies a 0.25-nat floor to the batch-and-spatial mean KL of each latent channel, so reducing a channel's KL below this threshold provides no further optimization benefit. Multiscale and feature-space terms are motivated by [8, 9], and free bits follow [10].

For a training anchor i with the requested species label, the encoder defines a diagonal-Gaussian posterior $q_\phi(z \mid x_i, y) = \mathcal{N}(\mu_i, \operatorname{diag}(\sigma_i^2))$. The corrected generation-time sampler applies temperature only to its stochastic residual:

$$z = \mu_i + T\sigma_i \odot \epsilon, \qquad T = 0.35, \quad \epsilon \sim \mathcal{N}(0, I). \tag{2}$$

This correction affects post-training sampling only; the training-time reparameterization and VAE checkpoint were unchanged. The train-only, content-safe posterior bank retained 256 Northern Cardinal, 247 Song Sparrow, and 256 American Robin anchors, and the three generation-seed VAE pools were regenerated from this bank. The method is conceptually related to mixture-of-posteriors priors such as VampPrior [11], but uses fixed encoded training examples rather than learned pseudo-inputs and does not modify the training objective.

3.3 Diffusion Model

4 Experimental Methodology

For the primary low-resource experiment, each newly initialized CRNN received 50 real training spectrograms per species, selected from distinct recording IDs. We crossed three real-data subset choices with three CRNN initialization seeds, yielding nine matched settings. Validation compared real-only training with adding 50, 100, or 200 generated spectrograms per species. The CRNN architecture, masking policy, optimizer-step budget, and validation/test splits were held constant; no pretrained CRNN was augmented. The augmentation amount was selected using validation macro-F1, while the primary outcome was the paired change in macro-F1 on the fixed held-out test set. Only the shared real-only baseline and each generator's selected condition were evaluated on the test set.

We also measured generator speed using five synchronized in-memory repeats of 600 spectrograms (200 per species) after warm-up, using FP32 precision, batch size 8, and the same RTX 4070 SUPER. Timing covered generation through the final normalized spectrogram tensor; loading, conversion, transfer, file output, waveform decoding, and metric computation were excluded.

5 Experimental Results

Classifier selection. The selected 404-k-parameter CRNN achieved a controlled three-seed validation macro-F1 of $90.45\% \pm 1.45\%$ and was used for the downstream augmentation experiments.

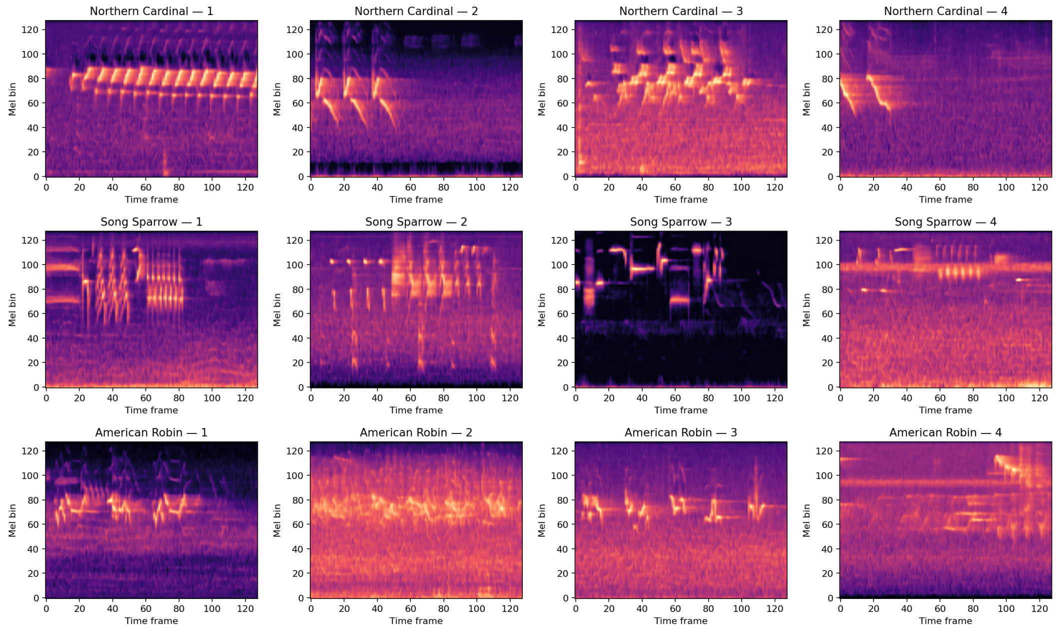

Figure 2: Existing VAE-v3 notebook posterior-anchor samples at temperature 0.35: four examples per species (48 samples were saved in total). Panels are independently color-normalized; spectrogram appearance alone does not establish waveform realism.

VAE diagnostics. The recorded test-set reconstruction MSE in the globally standardized log-mel domain was 0.1533 for V1, 0.0654 for V2, and 0.0281 for V3. V2 and V3 used deterministic posterior-mean reconstructions, whereas the historical V1 evaluation sampled from its posterior; the trend is therefore descriptive rather than a strictly matched deterministic comparison. V3 test MAE was 0.1261 and spectral convergence was 0.1794. MSE is an evaluation diagnostic rather than V3's complete training objective, and these input-to-reconstruction values are not generated-to-real distances.

For the separately verified seed-42 notebook pool (16 samples/species), the mean same-species generated-to-test nearest-neighbor pixel MSE was 0.5120 (SD 0.1998; median 0.4501; IQR 0.3878–0.6231); the leave-one-out real-to-real reference mean was 0.3350 (SD 0.1419; median 0.3185; IQR 0.2368–0.4154). The generated-to-test/real-to-real means by species were 0.4308/0.3237 for American Robin, 0.5579/0.3650 for Northern Cardinal, and 0.5474/0.3175 for Song Sparrow. These unpaired pixel-space summaries use 48 and 489 queries, respectively, and nearest-neighbor MSE can reward copying; they are neither perceptual-realism nor waveform-quality scores. This hash-recorded diagnostic used a separately reconstructed notebook pool (temperature 0.35; train-only posterior bank), rather than the canonical pools used downstream. Although both use standardized log-mels, 0.5120 and 0.0281 are not comparable because they summarize generated-to-nearest-test and input-to-own-reconstruction pairs, respectively. The Residual CNN remains a frozen VAE training teacher and supplies no independent VAE–Diffusion evaluation.

Generator-quality diagnostics. The selected CRNN was used as an evaluator of generated spectrograms, not as the downstream test model. Table 1 summarizes class compatibility, feature-space distribution, and exact-copy checks for the current pools. Values are means over the three generation seeds; feature rows additionally average the three species.

Across the current generation seeds, Diffusion reached $92.44\% \pm 0.86\%$ target-label accuracy and $92.42\% \pm 0.84\%$ macro-F1, compared with $96.67\% \pm 1.20\%$ and $96.66\% \pm 1.21\%$ for VAE-v3. VAE-v3 had lower feature Fréchet distance, higher feature precision, recall, density, and coverage, and lower pixel Wasserstein distance; Diffusion had a diversity ratio closer to one (1.0083 versus 0.8719). These current-pool, classifier-view diagnostics describe different quality dimensions and do not establish waveform realism, perceptual quality, or downstream augmentation utility.

Table 1: Generator-quality diagnostics for the current three-seed generated pools.

Metric	VAE-v3	Diffusion
Selected-CRNN target-label accuracy / macro-F1	96.67% / 96.66%	92.44% / 92.42%
Classifier-feature Fréchet distance	22.8950	41.2550
Feature precision / feature recall	0.8528 / 0.8972	0.5654 / 0.8207
Feature density / coverage / diversity ratio	0.6423 / 0.7419 / 0.8719	0.3114 / 0.4845 / 1.0083
Pixel Wasserstein / exact copies	0.0787 / 0	0.1916 / 0

Table 2: Validation selection and held-out test results for low-resource CRNN augmentation.

Added generated spectrograms per species	VAE-v3 validation macro-F1	Diffusion validation macro-F1	VAE-v3 test macro-F1	Diffusion test macro-F1
0 (real only)	82.24%	82.24%	84.75%	84.75%
50	84.53%	85.45%	–	–
100	85.88%	86.06%	–	–
200 (selected for both)	**86.47%**	**86.84%**	**87.48%**	**86.21%**

Low-resource augmentation. Table 2 summarizes the validation selection and held-out test results. Each condition used 50 real spectrograms per species. Validation values are means across the nine matched experimental settings and determined the selected augmentation amount. The shared real-only baseline and each generator's selected 200-generated condition were then evaluated on the fixed test set; dashes indicate conditions not evaluated on test. Diffusion had the higher validation macro-F1 at every nonzero augmentation amount, whereas VAE-v3 had the higher held-out test macro-F1. Mean held-out macro-F1 was 84.75% for real-only, 87.48% for VAE-v3, and 86.21% for Diffusion. The primary outcome, the paired change in held-out macro-F1, was $\Delta F1 = +2.72$ percentage points for VAE-v3 (7/9 positive blocks; descriptive block-bootstrap interval $+1.00$ to $+4.65$) and $+1.46$ points for Diffusion (6/9; $+0.15$ to $+2.89$).

Equal-count generator timing. The benchmark generated 600 fresh in-memory spectrograms (200 per species) per synchronized repeat after warm-up. Table 3 reports the complete request time, batch-average time per spectrogram, and throughput. The timing boundary ended at the final normalized spectrogram tensor and excluded model loading, conversion, transfer, file output, waveform decoding, and metric computation. Diffusion took $1348.09\times$ the VAE-v3 generator time for an equal-count request; measurements used FP32, batch size 8, five warm-up batches, five CUDA-synchronized repeats, and the same RTX 4070 SUPER.

Table 3: Generator-only timing comparison for matched fresh outputs.

Model	Mean time per 600 spectrograms (s)	Equivalent time per spectrogram (ms)	Throughput (spectrograms/s)
VAE-v3 (corrected sampler)	0.3058 ± 0.0096	0.510	1963.276
Diffusion (100-step DDIM)	412.3090 ± 2.8120	687.182	1.455

6 Discussion

In the simulated low-resource setting, both validation-selected 200-generated conditions improved mean held-out macro-F1 versus their matched real-only controls. The nine blocks reuse overlapping real subsets, so the intervals are descriptive and the largest tested ratio is not an estimated optimum. This result applies to newly initialized low-resource CRNNs, not the historical full-data CRNN. Reconstruction MSE rewards pixel proximity, the Residual CNN is coupled to V3 training and therefore restricted to its teacher role, and a closed-set independent CRNN cannot reject unknown or noisy inputs. Spectrogram and classifier metrics also do not replace listening or waveform-level evaluation [12]. VAE-v3 has higher CRNN target compatibility, lower macro-species Fréchet distance, and higher feature precision/coverage in these pools. Diffusion has a diversity ratio nearer one. Separately, Diffusion took $1348.09\times$ the VAE-v3 generator time under the recorded boundary.

7 Conclusion

BirdGEN now has corrected three-seed generator evidence, a completed matched low-resource CRNN study, and a matched timing comparison. Validation selected 200 additional generated spectrograms per species for both models; the mean paired macro-F1 changes were $+2.72$ points for VAE-v3 and $+1.46$ points for Diffusion. Quality diagnostics remain metric-specific rather than a composite winner, and the efficiency result is limited to generator-only spectrogram sampling on the recorded hardware.

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A Supplementary Reproducibility Notes

VAE-v3 used AdamW with learning rate 2×10^{-4}, batch size 24, a 20-epoch KL warm-up, and checkpoint selection by the complete validation objective. The portable pool generator now applies the notebook's sampling temperature 0.35 explicitly. The VAE checkpoint was not retrained. The existing posterior bank was filtered to 256 Northern Cardinal, 247 Song Sparrow, and 256 American Robin anchors, and the three canonical VAE pools were regenerated with the corrected sampler. The reported generated-to-test MSE instead uses a separate, hash-recorded seed-42 pool reconstructed with the original notebook, 16 samples per species, and the train-only posterior bank. This MSE package remains separate from the three-seed augmentation pools and does not replace their quality or downstream CRNN evidence.

For the equal-count timing study, each model generated 600 spectrograms (200 per species) in FP32 with batch size 8 on the same hardware. Five warm-up batches preceded five synchronized repeats. The sole timing boundary is generator sampling through the final normalized spectrogram tensor; model loading, warm-up, conversion, CPU transfer, file output, waveform decoding, plotting, and metric computation are excluded. The package preserves repeat seconds, throughput, hardware, precision, batch size, seeds, sampler settings, and the timing-boundary description.